Pulse — Testing and Quality Assurance Report

1. Introduction

This document is the consolidated testing and quality assurance report for Pulse. It covers the testing strategy applied across six sprints, the automated test suite that guards the scoring engine and app flows, the user testing programme run against a working build, the bugs found and their resolutions, and the risk register that drove decisions throughout the project.
Pulse was developed by a team of five between February and October 2026. The app ships with 305 passing automated tests, has been evaluated by two human participants against 15 user-test cases (plus additional domain-level cases), and has had every identified risk tracked, owned and mitigated.

2. Testing Strategy

2.1 Test levels

Level	Focus	Tools
Unit	Pure, deterministic logic — scoring formulas, clamping, mission selection, rating conversion.	Jest
Component / UI	Screen state, user feedback, form validation, loading behaviour.	React Native Testing Library + jest-expo
Integration / Service	Firebase data transformations, path ownership, deterministic conversation IDs.	Jest with mocked Firestore SDK
Simulation	14-week synthetic user behaviour across multiple scenarios, verifying scores stay within bounds.	Jest + custom seed generator
Manual device	Expo device behaviour, backend integration, session persistence, uploads, social	Physical iPhone & Android, iOS Simulator

Level	Focus	Tools
	updates, security rules.	
User testing	End-to-end flows exercised by independent participants.	Written test scripts + observer notes

2.2 Test field definitions

Field	Description
Feature / Area	The feature or app area being tested (e.g. Budget Tracker, Saving Plans).
Test Objective	What this test is intended to validate.
Key User Tasks / Scenarios	Actions and situation given to the participant.
Expected Outcome	What should happen if the feature works correctly.
Test Account	Account used to run the test.
Test Data / Preconditions	Budgets, scores, dates, records, user states.
Participant ID	Anonymous participant identifier (e.g. P01).
Device / OS	Device and OS used.
Task Success	Pass / Partial / Fail.
Completion Time	Time required.
Help Needed	None / Minor / Major.
Main Finding	Main observation, confusion, feedback or defect.
Severity	Low / Medium / High.
Action	Proposed change, follow-up task or decision.
Owner	Person responsible for the feature or follow-up.
Status	Not started / Testing / Open / Retest / Closed.

3. Automated Test Coverage

The test suite grew from zero in Sprint 2 to **305 passing tests across 5+ suites by Sprint 6**.

Suite	Tests	What it covers
scoringEngine	46	Rating conversion, domain score computation, balance score derivation, clamping.

Suite	Tests	What it covers
scoringSimulation	38	14 weeks of synthetic user behaviour across scenarios.
missionEngine	26	Domain weakness detection, mission generation with deduplication, pool integrity.
DailyMission component	18	Component rendering, completion toggling, Firestore persistence.
scoring integration	17	Firestore read/write for scoring.
Additional (Sprint 5–6)	~160	Auth retry, surplus calc edge cases, Dropdown disabled state, tutorial removal, navigation prop rename, consent flow, share flow.

The 14-week simulation tests validate the scoring algorithm over realistic timeframes. The reflection-only regression case is explicitly labelled *"Case 2: reflection only (no actions), fixed bug"* so the Sprint 5 fix is guarded against regression.
Two test suites (DailyMission.test.js, shareUi.smoke.test.js) currently fail to parse due to a pre-existing Jest/Babel config issue — tracked tech debt but does not affect the 305 passing tests.

4. Peer Code Review Process

21 formal Review ID (RID) documents were produced across the project. Reviews were conducted by a team member *other than* the developer, following a four-section template: Code review, UI review, Functionality check, Overall feedback.
Peer reviews caught issues including:

- A stray handleGlobalClick JSX attribute in the To-Do List.
- Missing error handling on Firestore writes.
- Unused state and empty functions in the Physical Activity Tracker.
- A production-reachable dummy-data seeder button that could overwrite real user data.
- Avatar upload path mismatch with Storage rules.
- Unguarded handleConsentComplete write that leaves the UI stuck on failure.

5. User Testing Programme

5.1 Approach

User testing was conducted in Sprint 6 using two participants (P01 and P02) across Android and iOS. Participants were given written scenarios and observed; completion time, help needed, and defects were recorded. Findings were then triaged.

5.2 Coverage

- **Entry Quiz** (4 tests) — all passed.
- **Physical Activity Tracker** (5 tests) — 1 pass, 3 partial, 1 blocker. 7 defects found.
- **Budget Tracker** (5 tests) — all passed. 1 low-severity finding.
- **Saving Plan** (5 tests) — all passed. 2 low-medium findings.
- **Sharing** (5 tests) — 4 passed, 1 blocked.
- **Daily Mission** (6 tests) — all passed.
- **Score System** (6 tests) — all passed.
- **To-Do List** (3 tests) — all passed after the 1/10/26 delete-option fix.
- **Social Posts** (4 tests) — all partial.

5.3 Outcome summary

Overall: **35 pass / 8 partial / 1 fail** reported to tutor in Sprint 6 week 2. The failure was the sharing-to-friend blocker — retestable rather than a true functional defect.

5.4 Bugs identified via user testing

Area	Finding	Severity	Status
All features	Keyboard covers input fields	Medium	Open
Authentication	Registration doesn't require real email verification	High	Open
Account settings	Cannot edit display name, avatar or password	Medium	Partially open — avatar blocked by Storage rules
Physical Activity	Strength display ambiguous ("15 × 12")	Medium	Open
Physical Activity	Manual per-set entry inefficient	Medium	Open
Physical Activity	No required-field indicators	Medium	Open
Physical Activity	Cannot delete saved activities	High	Fixed in physical_user_test_improvements
Saving Plan	Long-press to add savings undiscoverable	Medium	Open
Saving Plan	X button removes entire row	Low	Open
Social	Image upload not displaying app graphic	High	Open

Area	Finding	Severity	Status
	on posts		
Social	Long comments pushed off-screen	Low	Fixed in test_fix_sharing_issues
Social	Email of friend-to-add not visible	Low	Fixed in test_fix_sharing_issues
Social	Comment delete button positioned inconsistently	Medium	Fixed in test_fix_sharing_issues
Social	Post delete has confirmation; comment delete did not	Medium	Fixed in test_fix_sharing_issues
Spirituality	No share popup after mood check-in	Medium	Fixed in test_fix_sharing_issues
Todo	"Cleared" share template fired when tasks remained	Low	Fixed in test_fix_sharing_issues
Android	Mission pills visually distorted on Android	Medium	Fixed in test_fix_sharing_issues

6. Risk Management

The final register contains **27 identified risks**.

6.1 Risk register

ID	Risk	Cause	Category	L	I	Level	Mitigation	Owner
1	Team member sick	Illness	Human	2	3	Low	Share knowledge so others can cover.	Team
2	Stakeholder	Changing	Scope	3	4	Medium	Set clear expectati	Team

ID	Risk	Cause	Category	L	I	Level	Mitigation	Owner
	imposes sudden changes	requirements					ons and timeframes.	
3	Loss of contact with stakeholder	Stakeholder unavailable	Comms	2	4	Medium	Routine meetings; escalate to teaching team.	Yuting
4	Rogue merge requests	Merging without review	Technical	2	3	Low	Peer review every MR; git revert if needed.	Matthew
5	No team compensation	Stakeholder decision	Human	5	1	Low	Accepted as non-commercial student project.	—
6	ANU SSO down	External outage	Operational	1	5	Medium	Work offline until restored.	Team
7	Broken device	Equipment failure	Operational	2	3	Low	Push to remote twice/day.	Team
8	No HIVE access	Facility issue	Operational	1	1	Low	Online meeting or library.	Team
9	Backwards compatibility	Focus on newest OS only	Technical	1	2	Low	Tech stack with wide support.	Sajeevan
10	Data loss	Corruption, hardware	Technical	2	5	Medium	Regular pushes to GitLab.	Team
11	Weak testing /	Poor coverage	Technical	3	3	Medium	Two people	Team

ID	Risk	Cause	Category	L	I	Level	Mitigation	Owner
	bugs						write and verify tests.	
12	Poor communication	Forgetting to communicate	Comms	2	4	Medium	Announce files you'll touch in chat.	Team
13	Duplicate files	Poor file management	Technical	2	2	Medium	Refactor duplicates.	Team
14	Database leak	Poor security practice	Technical	1	1	Low	Encryption, MFA, access control for sensitive data.	Team
15	Scope creep	Stakeholder adds features	Scope	2	3	Medium	Prioritise core; manage expectations.	Yuting
16	Lack of codebase understanding	Team in silos	Comms	4	4	High	Share work; peer-review across members.	Team
17	Difficult merging	Infrequent commits	Technical	3	4	Medium	Communicate changes; test after each merge.	Team
18	Firebase paid tier	Stakeholder may not approve	Technical	4	2	Low	Fall back to text-only posts.	Sajeevan
19	Unfeasible feature	Suggestion too complex	Scope	4	3	Medium	Discuss feasibility;	Team

ID	Risk	Cause	Category	L	I	Level	Mitigation	Owner
	request						manage expectations.	
20	Data sync conflict	Concurrent writes	Technical	3	3	Medium	Timestamp/version resolution.	Team
21	UI layout overflow	Rich UI on small screens	Technical	3	2	Medium	Test multiple resolutions.	Team
22	Late change requests	Changes near end	Scope	3	4	Medium	Freeze scope at Sprint 5 end.	Team
23	Firebase quota exceeded	Heavy testing / seeding	Technical	3	4	Medium	Monitor usage; budget alerts; emulator.	Team
24	iOS photo permission denied	User taps "Don't Allow"	Technical	3	2	Low	Clear explanation; text-only fallback.	Sajeevan
25	Trademark / copyright	AI icon, unverified name	Scope	2	3	Medium	Document in handover; clearance before release.	Team
26	Incomplete handover docs	Time pressure	Comms	3	4	Medium	Dedicated Sprint 6 doc allocation.	Team
27	Improper system management after	Stakeholder unfamiliarity	Human	3	4	Medium	Clear setup guide + walkthrough.	Team

ID	Risk	Cause	Category	L	I	Level	Mitigation	Owner
	handover							

6.2 Highest risks and how they were addressed

- **Risk #16 (Lack of codebase understanding — High)** was neutralised via 21 formal peer reviews; team members used each other's code as templates.
- **Risks #2 and #15 (Stakeholder scope — Medium)** addressed by a feasibility-first evaluation framework.
- **Risk #22 (Late change requests)** addressed by Sprint 5 feature freeze, confirmed with stakeholder on 27/08/2026 and 03/09/2026.
- **Risk #23 (Firebase quota)** partially addressed by experimentalForceLongPolling + memoryLocalCache(), and by proactively flagging image-sharing cost to the stakeholder.

7. Quality Metrics Summary

Metric	Value
Total passing automated tests	305
Formal peer code reviews (RIDs)	21
User-test cases executed	40+ across 9 features
User-test pass rate (final)	35 pass / 8 partial / 1 blocked
Risks tracked	27
Critical risks neutralised	1 (Risk #16)
Fixes on test_fix_sharing_issues branch	12
Sprint 5 scoring bugs fixed	3

8. Outstanding Items for the Next Team

- Fix the three auth / Storage issues (K19, K20, K21) before any public release.
- Resolve the Jest/Babel config issue blocking DailyMission.test.js and shareUi.smoke.test.js.
- Investigate the regressed idempotent scoring on physical_user_test_improvements — do not merge as-is.
- Audit Firestore rules for new collection paths; tighten users/{uid} root doc read access.
- Set up CI pipeline — would have prevented the Sprint 5 white-screen incident.
